

COMPLEXITY AND MEMORY-BASED COMPARISON OF THE BRAIN ACTIVITY BETWEEN ADHD AND HEALTHY SUBJECTS WHILE PLAYING A SERIOUS GAME

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Received April 22, 2021

Accepted May 28, 2021

Published June 28, 2021

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Abstract

Attention deficit hyperactivity disorder (ADHD) is a mental health disorder that is very common among children and may last into their adulthood. It is known that ADHD affects the attention of patients due to problems with short-term memory. Therefore, analysis of attention and memory of these patients should come into consideration. In this study, the complexity and memory of Electroencephalogram (EEG) signals are analyzed to investigate the reduction in attention and memory of patients with ADHD compared to normal subjects while playing a serious game. To achieve this, the fractal dimension and sample entropy of EEG signals are analyzed to evaluate the alterations in the complexity of EEG signals. Moreover, the Hurst exponent of EEG signals for ADHD and non-ADHD subjects is calculated to discuss the memory of EEG signals. The results showed a smaller fractal dimension and sample entropy of EEG signals for patients with ADHD that reflects their lower attention. Besides, the Hurst exponent of EEG signals for these patients reflects their lower memory than normal subjects. Therefore, it can be concluded that the reductions of attention and memory in ADHD subjects are mapped on the reduction of complexity and memory of their EEG signals.

Keywords: Serious Game; ADHD; EEG Signals; Complexity; Fractal Dimension; Sample Entropy; Memory; Hurst Exponent.

1. INTRODUCTION

Since lack of attention in ADHD patients is associated with short-term memory, many works have been dedicated to analyzing patients' attention in different conditions. For this purpose, researchers have utilized different techniques to record and accordingly evaluate the brain reaction of these patients in various conditions. Several studies have evaluated the alterations of EEG signals,¹⁻⁴ MEG signals,^{5,6} MRI images,^{7,8} and PET images^{9,10} in patients with ADHD by utilizing different methods. Among these works, more focus has been paid on brain imaging of ADHD patients using the EEG technique.

Since patients with ADHD often show poor planning and organization of activities because of their reduced memory,¹¹ in this research, we analyze how the reduction in attention and memory for these patients affects the reduction of complexity and memory of their EEG signals. Although the analysis of the complexity of EEG signals in patients with ADHD has been conducted in some studies,¹²⁻¹⁴ however, no work has evaluated the effect of memory reduction for ADHD patients on the alterations of the memory of their EEG signals. Since EEG signals have self-affine structures,¹⁵ the fractal theory is a promising technique to evaluate the changes in their complexity and memory.

In general, fractals are identified as self-similar or self-affine objects. Fractal dimension is used to

quantify the complexity of these objects, where its larger values indicate greater complexity in objects.¹⁶ For years, many works have applied the fractal theory to analyze the complexity of various time series and images. Similarly, different biological and physiological time series (e.g. respiration signals,¹⁷ eye movements,¹⁸ MEG signals,¹⁹ HRV signals,²⁰ EMG signals,²¹ and DNA time series²²) have been analyzed using fractal theory.

Similarly, many works have been conducted to investigate the alterations in EEG signals using fractal theory. Works that analyzed the EEG signals' complexity in electrical,²³ visual,²⁴ olfactory,²⁵ and auditory²⁶ stimulation, emotional recognition,²⁷ classification of diseases,²⁸ detection of sleep inertia state,²⁹ and aging³⁰ can be mentioned.

Furthermore, the complexity of signals can be quantified using other techniques. In this study, sample entropy is chosen. Sample entropy is the indicator of the complexity of a dataset. The main reason for choosing sample entropy is that it is not sensitive to the length of data. Since recorded EEG signals from different subjects had different lengths, sample entropy is a suitable technique for our analysis. Therefore, the calculation of sample entropy validates the fractal analysis result. The sample entropy has been used extensively to evaluate the complexity of various types of physiological signals.³¹⁻³⁴ To be specific, the extensive application of sample entropy is highlighted in the analysis of EEG signals.³⁵⁻³⁷

Since we are interested in evaluating the memory of EEG signals for ADHD patients, we chose the Hurst exponent in this research. Hurst exponent is the indicator of memory of time series. In other words, it is a measure of predictability. Therefore, in this research, we investigate its suitability for linking the memory of EEG signals to the reduction in the memory of ADHD patients. Hurst exponent has been utilized by many researchers to investigate different physiological signals. We can also call the extensive application of the Hurst exponent in the analysis of EEG signals in different experiments.^{38–40}

Serious games or applied games are games that are designed mainly for a purpose other than pure entertainment. Due to the importance of serious games in the cognitive therapies field, particularly for patients with ADHD,^{41–43} analysis of the alterations of brain activity for ADHD subjects while playing these games is an important area of research. Therefore, in this research, we apply fractal theory and sample entropy to EEG signals to evaluate how the variations of their complex structures are correlated with the changes in the attention of ADHD subjects compared to non-ADHD subjects. Besides, calculating the Hurst exponent assists us in evaluating how the changes in the memory of EEG signals are correlated with the changes in the memory of ADHD and non-ADHD subjects.

In the following section, the applied methodology according to the fractal theory, sample entropy, and the Hurst exponent is discussed. Then, we will proceed with the data analysis, result, and discussion.

2. METHOD

In this paper, the alterations in brain reactions between ADHD and non-ADHD subjects during playing a serious game were analyzed. We were interested in investigating the effect of impaired attention and memory in ADHD subjects on the alterations of EEG signals. For this purpose, since EEG signals have complex structures, we employed fractal theory and quantified the EEG signals' complexity using the fractal dimension.

The fractal dimension can be calculated using various techniques. In this research, the box-counting algorithm was used. This algorithm segments a fractal object using boxes of the same size (ϕ). Then, N as the number of these boxes is calculated. The algorithm changes the size of these boxes in several steps. Finally, the fractal dimension is

calculated based on the number and size of boxes in each step⁴⁴:

$$F = \lim_{\phi \rightarrow 0} \frac{\log N(\phi)}{\log 1/\phi} \quad (1)$$

The fractal dimension in the general form is formulated as:

$$F_t = \lim_{\phi \rightarrow 0} \frac{1}{t-1} \frac{\log \sum_{i=1}^N p_i^t}{\log \phi} \quad (2)$$

In Eq. (2), t is the order of F , and p_i indicates the probability in the i th segment of the time series:

$$p_i = \lim_{l \rightarrow \infty} \frac{r_i}{l} \quad (3)$$

where r_i is the number of occurrences in the i th segment. The length of the time series is denoted by l .

We also quantified the EEG signals' complexity for ADHD and non-ADHD subjects using sample entropy. Sample entropy is a modification of approximate entropy that indicates the complexity of a dataset and is independent of data length. In fact, since EEG signals of different subjects had different lengths, sample entropy is applied to validate the fractal analysis results. Considering a signal in the form of " $(f(1), f(2), f(3), \dots, f(n))$ " with the embedding dimension of m and the tolerance of r , sample entropy (SamEntropy) is formulated as⁴⁵:

$$\text{SamEntropy} = -\log \frac{B}{C} \quad (4)$$

In Eq. (4), B and C are the number of vector pairs that $d[v_{m+1}(i), v_{m+1}(j)] < r$ and $d[v_m(i), v_m(j)] < r$ respectively, in which, $v_m(i)$ is defined using Eq. (5):

$$v(i) = [f(i), f(i+1), \dots, f(i+m-1)] \quad (5)$$

Moreover, we were also interested in investigating the reduction of memory in patients with ADHD. For this purpose, the memory of EEG signals between ADHD and non-ADHD participants was analyzed using the Hurst exponent. In other words, we wanted to examine whether the reduction of memory in ADHD patients is mapped in the reduction of memory of EEG signals.

Hurst exponent indicates the memory of time series. For a time series, Hurst exponent (H) and the fractal dimension (F) have the following relation⁴⁶:

$$H = 2 - F \quad (6)$$

Hurst exponent can have any value between 0 and 1, when 0.5 indicates a completely random process (Brownian motion). If the value of the Hurst exponent deviates from 0.5 toward 0 or 1, it indicates the higher memory of the process. For example, a time series with the Hurst exponent of 0.8 has higher memory than a time series with the Hurst exponent of 0.6. In another example, a time series with the Hurst exponent of 0.3 has higher memory than a time series with the Hurst exponent of 0.4, since 0.4 is closer to 0.5.

Therefore, the fractal dimension and sample entropy of EEG signals for ADHD and non-ADHD subjects were computed to evaluate the differences in their complexity. Furthermore, the Hurst exponent of EEG signals was calculated to relate the reduction of memory of ADHD subjects to the memory of their EEG signals.

3. DATABASE AND ANALYSIS

In this study, we used a database that has been prepared by Alchalabi *et al.*⁴⁷ They designed a serious game named FOCUS using a unity game engine. The developed game could be controlled using an EEG device and keyboard. In EEG-based controlling, the game challenged the player to move an avatar by focusing and using mental commands to collect all the cubical pickups in the shortest possible time (Fig. 1). To record the EEG signals, an Emotiv EPOC EEG device (Emotiv, USA) with 14 recording electrodes and two reference electrodes has been used. The placement of EEG electrodes was based on the 10–20 system.

The experiment has been conducted on five healthy subjects (non-ADHD) and four participants with ADHD. Participants were asked to play the game in two sessions, one time with keyboard

arrows and another time using the EEG device. In both sessions, their EEG data has been recorded at 128 Hz. The raw EEG data were extracted and recorded using python scripts.

Alchalabi *et al.* divided the original EEG signals into different frequency bands.⁴⁷ Since the gamma band (with a frequency of above 30 Hz) is related to problem-solving and memory,⁴⁹ in this research, we used gamma-band EEG signals for our analysis.

We computed the fractal dimension, sample entropy, and Hurst exponent for different samples of gamma bands from EEG signals for ADHD and non-ADHD participants. For this purpose, a set of codes that have been written in MATLAB (MathWorks, USA) was used according to the discussed methodology. For the computation of the sample entropy, the values of embedding dimension and tolerance respectively were chosen as 2 and $0.2 \times \text{Standard deviation of data}$. For comparison, four subjects from each group (ADHD and non-ADHD) were considered in our analysis.

Using Student's t-test ($\alpha = 0.05$), pairwise comparisons were performed on the calculated fractal dimension, sample entropy, and Hurst exponent of EEG signals to check the significance of their changes between ADHD and non-ADHD subjects.

4. RESULTS

Figures 2 and 3 show the fractal dimension and sample entropy of EEG signals for subjects while controlling the game using their brains. As illustrated in Fig. 2, the EEG signals of non-ADHD subjects had a greater fractal dimension. Therefore, it can be stated that the EEG signals' complexity was greater for non-ADHD compared to ADHD subjects. The result of the t-test ($p = 0.004$) demonstrates that



Fig. 1 The game environment.⁴⁸

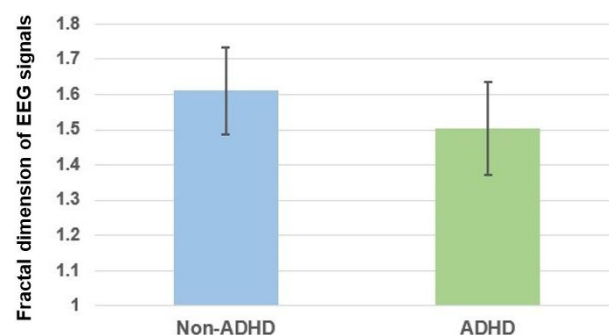


Fig. 2 The fractal dimension of EEG signals while playing the game using the brain. Error bars indicate standard deviation.

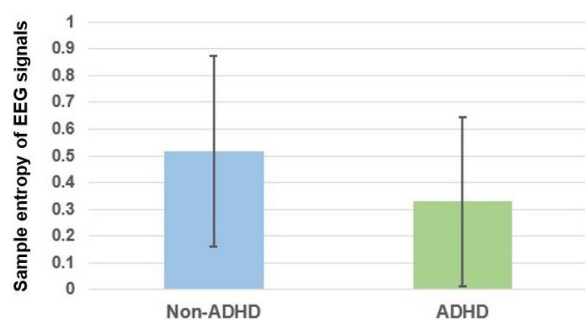


Fig. 3 The sample entropy of EEG signals while playing the game using the brain. Error bars indicate standard deviation.

the EEG signals' complexity for non-ADHD subjects is significantly higher than the one for ADHD subjects.

By referring to Fig. 3, it can be understood that the analysis of sample entropy shows similar results with the results of fractal analysis; the EEG signals for non-ADHD subjects show greater complexity than the one for ADHD subjects. Besides, the result of the t-test ($p = 0.0379$) demonstrates that the EEG signals' complexity for non-ADHD subjects is significantly higher than the one for ADHD subjects. Therefore, the analysis of sample entropy verifies the results of fractal analysis. As previously mentioned, ADHD causes less attention in subjects. Therefore, due to the less attention, the complexity of their EEG signals is lower compared to subjects with normal brains while playing the game.

Figures 4 and 5 show the fractal dimension and sample entropy of EEG signals for subjects while controlling the game using a keyboard. As illustrated in Fig. 4, the EEG signals of non-ADHD subjects had a greater fractal dimension, and therefore, it states that the complexity of EEG signals was greater for non-ADHD compared to ADHD subjects. The result of the t-test ($p = 0.00001$) demonstrates that the EEG signals' complexity for non-ADHD subjects is significantly higher than the EEG signals' complexity for ADHD subjects.

By referring to Fig. 5, it can be understood that the analysis of sample entropy shows similar results with the results of fractal analysis; the EEG signals for non-ADHD subjects show greater complexity than the one for ADHD subjects. Besides, the result of the t-test ($p = 0.00001$) demonstrates that the EEG signals' complexity for non-ADHD subjects is significantly higher than the EEG signals' complexity for ADHD subjects. Therefore, the analysis of sample entropy verifies the results of fractal analysis. Therefore, by referring to Figs. 2–5, it can

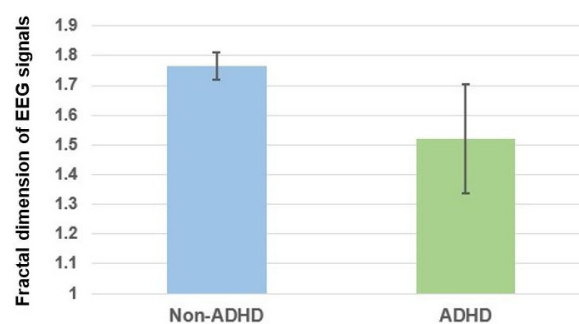


Fig. 4 The fractal dimension of EEG signals while playing the game using a keyboard. Error bars indicate standard deviation.

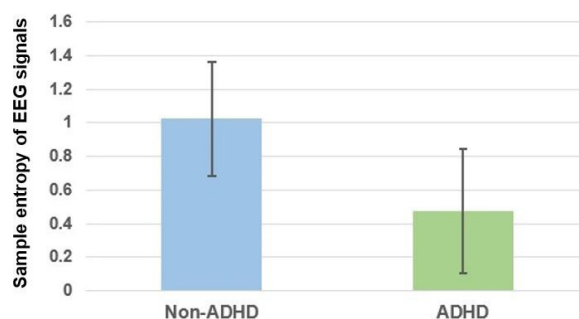


Fig. 5 The sample entropy of EEG signals while playing the game using a keyboard. Error bars indicate standard deviation.

be recognized that EEG signals of non-ADHD subjects in cases of brain and keyboard-based controlling of games show higher complexity than ADHD subjects.

As previously mentioned, since ADHD symptom affects the memory of subjects, we also analyzed the Hurst exponent of EEG signals to see how their impaired memory affects the memory of EEG signals.

Figures 6 and 7 show the Hurst exponent of EEG signals during the brain and keyboard-based controlling of the game, respectively. As illustrated in both figures, the Hurst exponent of EEG signals for ADHD subjects obtained the values which are very close to 0.5. This Hurst exponent of 0.05 reflects the Brownian motion in which no memory is associated with the signal. Therefore, it can be stated that the EEG signals of ADHD subjects have very low memory. On the other hand, since the Hurst exponent of EEG signals for non-ADHD subjects is not close to 0.5, therefore, it can be declared that their EEG signals have higher memory than the ones for ADHD subjects. By referring to the concept of impaired memory in ADHD subjects, it can be seen that the reduction in memory has reflected on the decrease

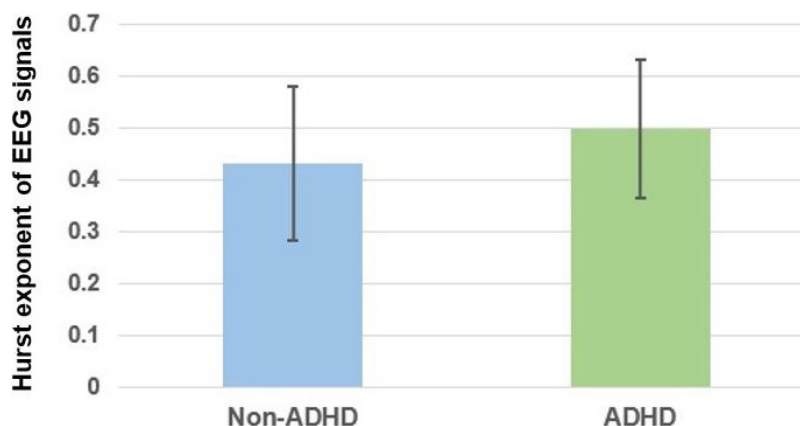


Fig. 6 The Hurst exponent of EEG signals during playing the game using brain. Error bars indicate standard deviation.

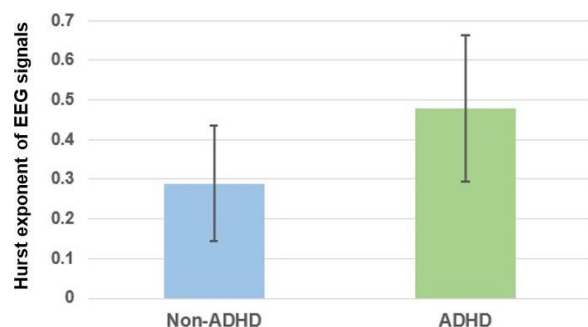


Fig. 7 The Hurst exponent of EEG signals while playing the game using a keyboard. Error bars indicate standard deviation.

in memory of their EEG signals compared to non-ADHD subjects.

In general, it can be said that ADHD caused the reduction of complexity and memory in the EEG signals of patients compared to healthy subjects.

5. DISCUSSION

In this paper, we analyzed the alterations in the complexity of EEG signals between normal and ADHD subjects while playing a serious game. For this purpose, we utilized a database that presents EEG signals of subjects while controlling the game using a keyboard and, in another approach, using their brains. To analyze the reduction in attention for ADHD subjects, the fractal exponent and sample entropy of their EEG signals were analyzed. The results of the fractal analysis indicated the significantly greater complexity of EEG signals for healthy subjects compared to ADHD subjects. Besides, the analysis of the sample entropy of EEG signals also showed similar results with the

result of fractal analysis in which the EEG signals for subjects with ADHD have significantly lower complexity than non-ADHD subjects. Therefore, based on the result of the analysis, it can be concluded that the lower attention of ADHD subjects is mapped on the lower complexity of their EEG signals. This conclusion is in line with the presented results in Ref. 50, which indicate when the brain is engaged with greater processing, the EEG signals' complexity increases. Therefore, since people with ADHD pay less attention to the changes around them, the brain has lower processing, and therefore, their EEG signals have lower complexity than healthy subjects.

Besides, we also analyzed the Hurst exponent of EEG signals for ADHD and non-ADHD participants. The results revealed that EEG signals for ADHD subjects have significantly lower memory than non-ADHD subjects. Since impaired memory is one of the symptoms of ADHD, it can be seen that the reduction in the memory of ADHD subjects is mapped on the decrease in memory of their EEG signals compared to normal subjects. This inference is in line with the presented result in Ref. 51, which indicates that when the brain is engaged with the processing of external stimuli, the memory of EEG signals increases. Similarly, when subjects with ADHD played the focus game, their memory has been increased less than the increase in the memory of normal subjects, which is mapped on the lower memory of their EEG signals compared to healthy subjects.

The conducted analysis in this research helped us to link the concept of attention and memory of ADHD subjects to complexity and memory of EEG signals, respectively. This is the first study that

analyzed this coupling. Here, we cannot propose a direct relationship between the brain's memory (and attention) and the memory (and complexity) of EEG signals. However, based on our analysis results, the fractal theory is a suitable technique to decode this correlation. The suitability of the fractal theory for this analysis can be understood by referring to its nature. It is known that the Hurst exponent is a measure of predictability. In Ref. 52, it was shown that the Hurst exponent could be used to forecast EEG signals based on their past fluctuations. Therefore, linking the memory of the brain to the memory of EEG signals is reasonable.

The conducted analysis in this study can be extended to other brain disorders. For instance, researchers can conduct a similar analysis for patients with Alzheimer's to evaluate whether the reduction of memory in these patients is mapped on the decrease in memory of EEG signals. Overall, it can be said that all these analyses help scientists for a better understanding of brain activities for patients with different brain disorders.

ACKNOWLEDGMENT

This work was supported in part by the project (2021/2205), Grant Agency of Excellence, University of Hradec Kralove, Faculty of Informatics and Management, Czech Republic.

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